何锦程 DAVID

Homepage (Portfolio)

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- C++ Python C# Python
- DirectX, OpenGL, Pytorch3D, Pytorch
- Blender Houdini
- Unreal Engine, Unity

MASTER OF SCIENCE | GAMES ENGINEERING
2024.9 - 2026.1
BACHELOR OF SCIENCE | COMPUTER SCIENCE
2020.9 - 2024.7

WARWICK UNIVERSITY | UNITED KINGDOM
Advanced Computer Graphics, Machine Learning, Game Design
HONGKONG BAPTIST UNIVERSITY | ZHUHAI, CHINA
Data Structure and Algorithm, operation system, Deep Learning, Computational Theory

RESEARCH ASSOCIATE

Hong Kong Polytechnic University | Computing Department | Hong Kong, China
2025.12 – PRESENT

HUAWEI LIGHTWEIGHT DYNAMIC NEURAL RENDERING PROJECT : Encoded base materials with **neural textures** with two-stage training: first learning materials along with static/dynamic GI, then jointly optimizing textures and light probes on an interpolation grid over moving-object positions to fit position-dependent shadows, improving training efficiency.

MATERIAL AND DYNAMIC LIGHTING DECOUPLING ARCHITECTURE: Stored spherical harmonic (SH) lighting coefficients in a triplane. Scene conditions such as object poses and switch states are encoded and fed into per-plane Conv2D networks that predict lighting residuals, enabling real-time prediction of dynamic lighting changes. Reduced storage by 10× compared with voxel probes. Lighting residuals are pre-baked at grid nodes across the condition space and interpolated at runtime, so no convolution runs on the device.

DUAL-MLP ARCHITECTURE: A trunk-head network with separate diffuse and specular branches, decoupling diffuse from high-frequency specular prediction to improve the expressiveness of each. Used rasterization G-buffer data and shading models to strengthen supervision while further reducing computational cost. Achieved a stable 25 dB PSNR at 90+ FPS on complex dynamic-GI scenes on a Huawei Mate 70 Pro.

3D SCENE CONSTRUCTION & DCC AUTOMATED DATA PIPELINE: Utilized Blender to build high-fidelity 3D scenes. Implemented the automated data pipeline bridging traditional DCC assets and neural rendering by automatically extract G-buffer features (BaseColor, Normals, etc.)

BACKEND PIPELINE ENGINEER

WarpEngine | Algorithm & Technology Dept. |
2026.7 – PRESENT

3D SCENE RECONSTRUCTION: Independently contructed pipeline that reconstructs editable 3D scenes from monocular video. It covers object-level reconstruction, scale and geometry calibration, dynamic human recovery, and physically valid scene assembly, and produces scene data for world-model training (see Projects).

AUTOMATED SCENE SCANNING & DATA COLLECTION: Develop and maintain an automated scene-scanning and batch-export toolchain for game-asset workflows and model training. The toolchain covers in-scene traversal sampling, gameplay capture, and batch export end to end; fully scripted processing replaces manual steps and supports large-scale training data collection.

DISTRIBUTED RENDERING CLUSTER: Maintain the central dispatch service for a multi-node, multi-GPU rendering cluster, using Supervisor to manage task dispatch and status synchronization across render nodes. Each render job runs as three sequential Docker containers (asset sync, core rendering, post-processing) to keep throughput stable under high concurrency.

MONOCULAR VIDEO 3D SCENE RECONSTRUCTION

2026.8 – PRESENT

Engineered this 3D scene production pipeline for world-model training end to end. The system reconstructs object-level editable, metric-scale indoor and outdoor scenes from monocular real-world video, and supports dynamic humans and basic physical-simulation constraints. It has been validated end to end on dynamic indoor and outdoor scenes with camera motion and moving subjects.

SYSTEM ARCHITECTURE: Geometric measurement determines object position and orientation, generative models complete the shapes of unseen regions, and semantic priors set absolute scale. Multiple models are combined for complementary recall so that no objects are missed. Static objects use single-image 3D generation (sam3d object) and humans use 4D mesh recovery. Indoor and outdoor scenes share one backbone and differ only in how the scene shell is built.

SCALE & GEOMETRIC CALIBRATION: Point clouds back-projected from multi-frame masks are fused and fitted with gravity-aligned oriented bounding boxes, which anchor the placement of generated meshes.

Two-level deduplication over 2D masks and 3D voxels removes cross-category duplicate detections. To handle monocular errors, global scale is calibrated with VLM semantic priors, outlier frames are rejected through cross-frame depth checks, and gravity and orientation alignment are unified, which removes scale bias and coordinate-frame misalignment.

DYNAMIC HUMANS & PHYSICS: A video-based 4D human pipeline (segmentation and tracking, diffusion-based occlusion completion, parametric body reconstruction) that outputs temporally consistent human sequences under occlusion, aligned to the reconstructed scene's coordinate frame. The scene shell (indoor walls, outdoor ground) serves as static colliders, and rigid-body simulation settles objects and resolves interpenetration

ENGINEERING & PRODUCTIONIZATION: An automated pipeline with a single-entry point, checkpoint resumption, status queries, and quality gates between stages. Decisions are split by whether a machine can make them reliably: the pipeline runs automatically up to each quality gate and asks for human confirmation only where automated judgment is unreliable.

Python, PyTorch, SAM 3, SAM 3D Objects, SAM 3D Body, Depth Anything 3, VGGT, Qwen3-VL, SpatialLM

RESEARCH ASSISTANT INTERN

Hong Kong University of Science and Technology | Guangzhou, China

2025.7 – 2025.9

PCG Generation Visualization: Responsible for the Unity-based VR visualization of Procedural Content Generation (PCG) projects.

IT INTERN

Brilliance BMW Automobility | Shenyang, China

2022.7 – 2022.9

Participated in the update of the employee application's shuttle route system. Executed end-to-end functional test cases to verify the accuracy of queries and display functions for all new routes and schedules.

HOUDINI-BASED PCG CITY GENERATOR (MASTER'S DISSERTATION) *BASED ON HOUDINI*

Developed a Procedural Content Generation (PCG) city generator supporting custom urban zoning and road networks.
Implemented a visual parameter panel for zoning and integrated external asset import functionality Youtube demo link

Basic Path Tracer

SUPERVISOR: PROF. THOMAS BASHFORD-ROGERS | NOV 2024 - JAN 2025

Developed core rendering features, including Physically Based Rendering (PBR) and Path Tracing for global illumination.
Implemented the BVH, Importance Sampling, and Adaptive Sampling. Demo Description

EXTREME SPORTS GAME

This project is a procedural level generation system for a mountain biking game inspired by 'Riders Republic'.
My primary role was designing the pipeline for realistic terrain generation and an advanced procedural pathfinding system, utilizing Houdini for generation and Unreal Engine 5 for real-time gameplay. Demo Description / Youtube demo link

2D Pixel-Art Dual-Stick Shooter Game(Game Jam Project) Indie Game development

SPET 2024

Developed a 2D twin-stick shooter using C++ and DirectX.
Created skeletal animations in Blender and produced custom pixel-art sequences.
Developed custom NPC data structures, data-driven maps, collision detection, and animation systems.
Demo Description / Youtube demo link

INTERESTS & HOBBY

Indie game development, Open World PCG, Computer Graphics, Weight lifting, Jazz